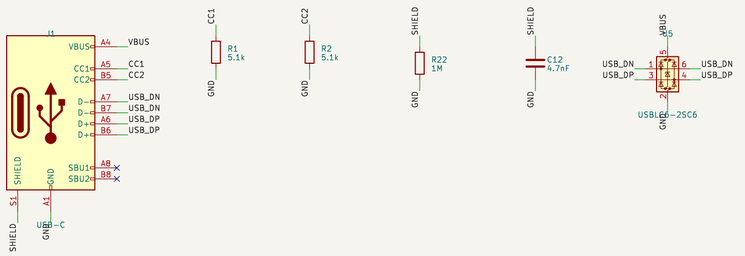
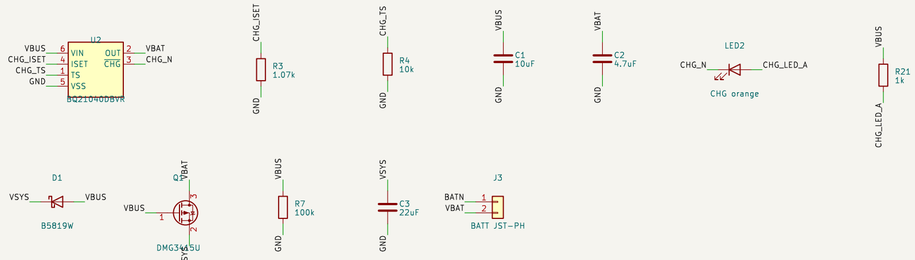
USB-C input and ESD



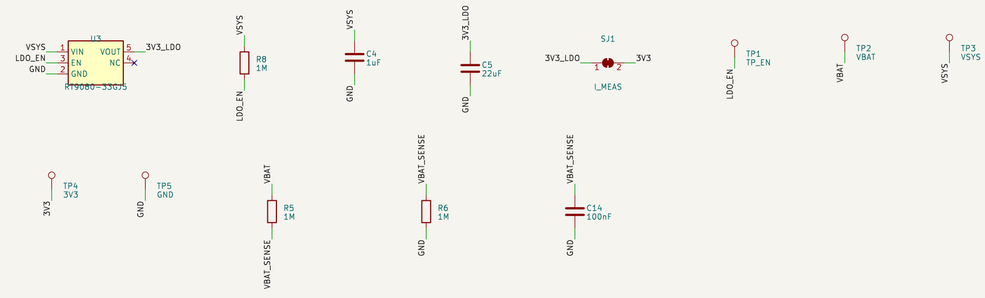
Li-ion charger and load-sharing power path



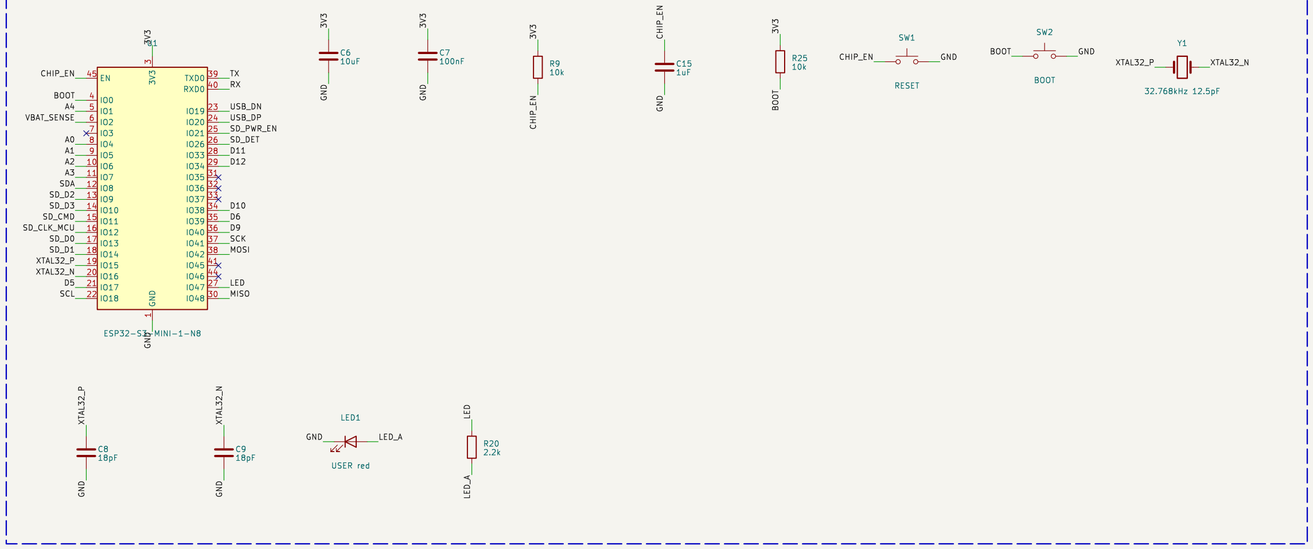
Optional cell protection (DNP; SJ2 bypass fitted)



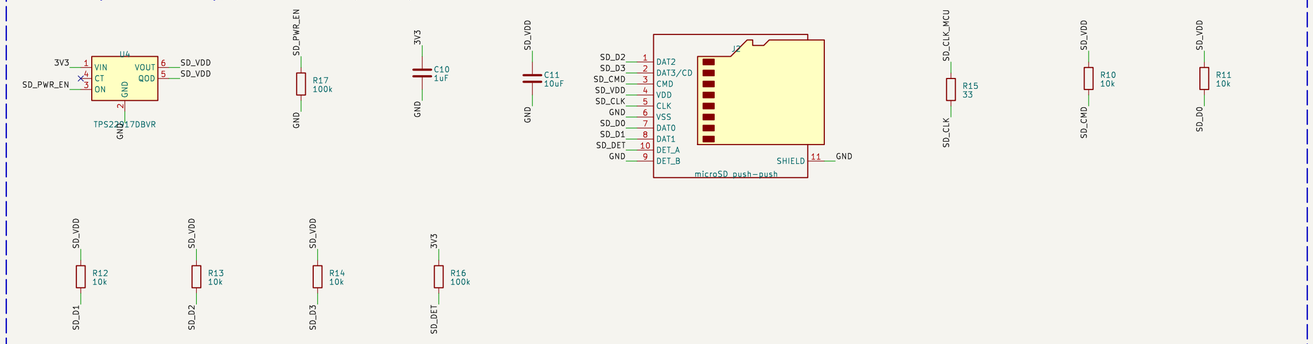
3.3 V regulator, battery sense, test hooks (E6/E7)



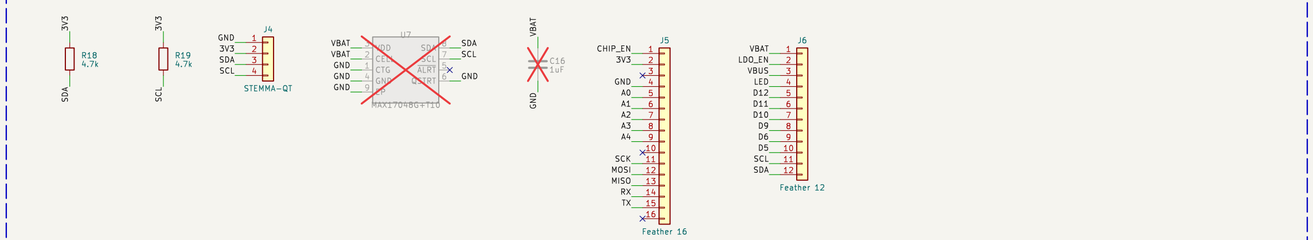
ESP32-S3-MINI-1-N8, reset/boot, 32 kHz crystal, LED



microSD (4-bit SDMMC), power switch, pull-ups



I2C / STEMMA-QT, optional fuel gauge, Feather headers



ZB-1 rev 0.1 — generated from scripts/zb1_design.py (single source of truth). Edit the script, not this file.
DNP by default: U6 + Q2 + R23/R24/C13 (cell protection; SJ2 bridges BATN–GND), U7 + C16 (fuel gauge).
J3 polarity: pin 1 = BAT- (black), pin 2 = BAT+ (red). CHECK YOUR CELL — there is no polarity standard.
SJ1: cut to insert an ammeter in the 3V3 rail (E6). TP1 / Feather EN pulls LDO_EN low to cut power (E7).

Design only — not fabricated or tested. See ../README.md
ESP32-S3-MINI-1-N8 | BQ21040 LiPo charger | RT9080 LDO | 4-bit microSD
ZEMIZDAT project (github.com/danindiana/zemizdat)

Sheet: /
File: zb1.kicad_sch

Title: ZEMIZDAT ZB-1 development board

Size: A2	Date: 2026-10-19	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1